

Mafalda Ramôa

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Research Experience

- June 2026 – **Postdoctoral Associate**
Current Virginia Tech Center for Quantum Information Science and Engineering, USA
Economou-Barnes Group
- August 2023 – **Research Scholar**
June 2026 Virginia Tech Center for Quantum Information Science and Engineering, USA
Economou-Barnes Group

Education

- 2022 – 2026 **PhD in Computer Science**
Advisors: Sophia Economou (Virginia Tech), Luís Paulo Santos (University of Minho), Ernesto Galvão (INL)
Project: On the Quest for Shallow, Accurate, and Cost-Efficient Ansätze for Variational Quantum Algorithms
- 2016 – 2021 **Integrated Masters in Engineering Physics - Physics of Information**
University of Minho
Final Average: 18/20
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- Master Thesis: [Ansätze for Noisy Variational Quantum Eigensolvers](#)
Grade: 19/20
Advisors: Ernesto Galvão (INL), Mikhail Vasilevskiy (University of Minho)
Informal supervision: Raffaele Santagati (Boehringer Ingelheim)

Publications

- 2026 *An efficient algorithm for approximate shadow Hamiltonian simulation*
A. Chakraborty, B. Sambasivam, K. Shiralì, H. Nelson, **M. Ramôa**, S. E. Economou, E. Barnes
[arXiv:2607.11882](https://arxiv.org/abs/2607.11882)

- 2026 *Continuous-variable ADAPT-VQE for bosonic lattice models*
D. Athanasakos, G. Tejedor-García, J. Y. Araz, **M. Ramôa**, B. Sambasivam, S. E. Economou, F. Ringer
arXiv:2606.05297
- 2026 *Co-Designed Adaptive Quantum State Preparation Protocols*
M. Ramôa, L. P. Santos, N. J. Mayhall, E. Barnes, S. E. Economou
arXiv:2601.20681
- 2026 *The cost of quantum algorithms for biochemistry: A case study in metaphosphate hydrolysis*
R. LaRose, A. Bidart, B. DalFavero, S. E. Economou, J. Wayne Mullinax, **M. Ramôa**, J. Rowland, B. Rubenstein, N. PD Sawaya, P. Vaish, G. M. Rotskoff, N. M. Tubman
arXiv:2601.19059
- 2025 *Strategies for Overcoming Gradient Troughs in the ADAPT-VQE Algorithm*
J. Stadelmann, J. Übelher, **M. Ramôa**, B. Sambasivam, S. E. Economou
arXiv:2512.25004
- 2024 *Reducing Measurement Costs by Recycling the Hessian in Adaptive Variational Quantum Algorithms*
M. Ramôa, L. P. Santos, N. J. Mayhall, E. Barnes, S. E. Economou
Quantum Sci. Technol. 10 (2024) 015031
- 2024 *Reducing the Resources Required by ADAPT-VQE Using Coupled Exchange Operators and Improved Subroutines*
M. Ramôa, P. G. Anastasiou, L. P. Santos, N. J. Mayhall, E. Barnes, S. E. Economou
npj Quantum Inf 11, 86 (2025)

Conferences and Talks

- 23 Apr 2025 Talk at Brown University
State-of-the-Art of Adaptive Variational Quantum Algorithms
- 28 Nov 2025 Talk at Rio de Janeiro Federal University (UFRJ)
Near-Term Quantum Algorithms for Quantum Chemistry: Progress and Prospects
- 1-8 Aug 2024 Poster Presentation at the Quantum Simulation (QSim) 2025 Summer School and Conference, New York, USA
M. Ramôa, P. G. Anastasiou, L. P. Santos, N. J. Mayhall, E. Barnes, S. E. Economou,
Status and Prospects of Variational Quantum Algorithms

16-21 Mar 2025	Talk at the APS Global Physics Summit, California, USA M. Ramôa, P. G. Anastasiou, L. P. Santos, N. J. Mayhall, E. Barnes, S. E. Economou, <i>Reducing the Resources Required by ADAPT-VQE Using Coupled Exchange Operators and Improved Subroutines</i>
25-29 Nov 2024	Talk at the Quantum Techniques in Machine Learning (QTML) Conference, Melbourne, Australia M. Ramôa, L. P. Santos, N. J. Mayhall, E. Barnes, S. E. Economou, <i>Reducing Measurement Costs by Recycling the Hessian in Adaptive Variational Quantum Algorithms</i>
09-16 Aug 2024	Poster Presentation at the Quantum Simulation (QSim) 2024 Summer School and Conference, Rhode Island, USA M. Ramôa, P. G. Anastasiou, L. P. Santos, N. J. Mayhall, E. Barnes, S. E. Economou, <i>Improving Adaptive Variational Quantum Algorithms</i>
24-28 Jul 2023	Poster Presentation at the Theory of Quantum Computation, Communication and Cryptography (TQC) Conference, Aveiro, Portugal M. Ramôa, R. Santagati and E. Galvão, <i>ADAPT-VQE: Impact of Noise, Importance of Symmetries, and Circuit Depth Reduction via Operator Removal</i>

Awards and Recognition

2025	Outstanding Reviewer Award – Quantum Science and Technology, IOP Publishing
2025	IOP Publishing Trusted Reviewer Status
2024	”R&D@USA” Scholarship (Luso-American Development Foundation)
2020/21	Excellence Scholarship Prize (University of Minho) <i>Awarded to the best student of each degree</i>
2020/21	”New Talents in Quantum Technologies” Scholarship (Gulbenkian Foundation)
2019/20	Excellence Scholarship Prize (University of Minho)
2017/18	Excellence Scholarship Prize (University of Minho)

Peer Review

Reviewing activity for Nature Physics, Nature Communications, Nature Reviews, Communications Physics, PRX Quantum, Quantum Science and Technology, Quantum, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, Physica Scripta, Digital Discovery, PRRResearch, Journal of Chemical Physics, Journal of Chemical Theory and Computation, AVS quantum science

Outreach, Events and Teaching

5-17 Oct 2026	Tutor at International Autumn School in Quantum Algorithms for Many-Body Problems (IASQA) , Cargèse (France)
27-30 Aug 2026	Tutor at C ² QA QIST Summer School, Brookhaven National Laboratory and Virginia Tech Online Summer School on Quantum Computing for High School Students
8-9 Nov 2025	VTQ Hackathon Co-Organizer (total 2 organizers) Overnight hackathon event at the Virginia Tech Blacksburg campus with 100 participants. Developed the hackathon prompt (about quantum amplitude estimation and circuit transpilation), managed event preparation and participant support in person and online, and judged the solutions
Sept-Nov 2025	Qiskit Fall Fest Event Organizer
28-21 Jul 2025	Tutor at C ² QA QISE Summer School, Brookhaven National Laboratory and Virginia Tech Online Summer School on Quantum Computing for High School Students
Mar 2025-Current	Supervisor of a student at the Lake Zurich High School (USA) Project on reducing the impact of noise on ADAPT-VQE operator selection using quantum error mitigation techniques
Nov 2024-Current	Supervisor of two students for Diploma Thesis at HTL Bregenz (Austria) Project on the choice of operator ordering in adaptive ansätze for VQE (Resulting paper)
Sept 2024-Sept 2025	Supervisor of an MSc student at the University of Minho (Portugal) Project on the viability and costs of fault-tolerant VQAs
Sept 2024-Current	Supervisor of three students at the Thomas Jefferson High School (USA) Project on the viability of a spin-preserving variant of ADAPT-VQE
10 Jul 2024	Tutor at C-Tech ² , Virginia Tech, USA In-person Summer School on Quantum Computing for High School Students
5-8 Aug 2024	Tutor at C ² QA QIST Summer School, Brookhaven National Laboratory and Virginia Tech Online Summer School on Quantum Computing for High School Students
Jun-Sept 2024	Supervisor of a BSc student at the Longwood University (USA) Project on VQE ansatz design for condensed matter Hamiltonians

Technical skills

Programming languages

Proficient: Python

Familiar: C, Assembly, R, Java, Haskell, Matlab

Software

Qiskit, Cirq, OpenFermion, PyZX, $\LaTeX$

Languages

Native: Portuguese

Fluent: English

Basic knowledge: French, Spanish, Italian